静态路由 Route Static

实验目的

- 了解路由器的作用
- 熟悉路由器的基本结构
- 熟悉路由器的基本操作
- 掌握静态路由的配置
- 进一步熟悉 eNSP 的使用
- 进一步熟悉利用抓包工具分析网络的方法

实验设备

- eNSP 网络模拟平台
- 路由器 AR2220 3 台
- 交换机 S5700（可选）
- 主机 PC（可选）

前置知识

- 人工路由
- 由网络管理员手动配置，用于明确指定路由器将数据包转发到目标网络的路径
- 不会自动学习或更新；网络拓扑变化时必须人工修改
- 特殊静态路由：0.0.0.0/0；默认路由
- 适合小型网络、边缘网络、出口默认路由、链路数量少且拓扑稳定的环境
- 中大型网络或链路经常变化的网络，一般会改用 OSPF 等动态路由协议

表 1 实验用到的主要命令

命令	**说明**
system-view	进入系统视图（system view）
sysname xxx	更改设备命令提示符
undo info-center enable	关闭回显信息
interface type slot/sub/port	选择端口
undo shutdowm	开启端口；默认是打开的
ip route-static NETWORK MASK NEXT-HOP	配置静态路由
display interface brief	查看所有端口概要信息
display interface GigabitEthernet 0/0/0	查看特定端口信息
display ip routing-table	查看路由表

命令	说明
display ip routing-table protocol static	仅仅查看静态路由
display version	显示设备版本号
undo	删除配置命令
quit	返回上一级视图
return	返回用户视图
save	保存配置（用户模式 user view）

实验步骤

路由器基本操作

1. 开启
2. 进入系统视图
3. 关闭回显信息
4. 更改主机名/命令提示符
5. 选择端口
6. 配置 IP 地址
7. 其它操作

完整操作

1. 创建网络拓扑

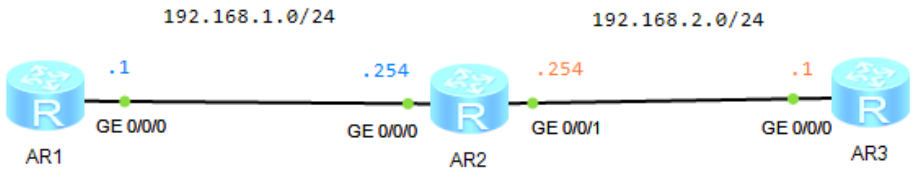


图 1 网络拓扑

2. 配置路由器 IP 地址

AR1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname AR1
[AR1]undo info-center enable
Info: Information center is disabled.
[AR1]interface g0/0/0
```

```
[AR1-GigabitEthernet0/0/0]ip address 192.168.1.1 24
[AR1-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[AR1-GigabitEthernet0/0/0]quit
[AR1]
```

AR3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname AR3
[AR3]undo info-center enable
Info: Information center is disabled.
[AR3]interface g0/0/0
[AR3-GigabitEthernet0/0/0]ip address 192.168.2.1 24
[AR3-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[AR3-GigabitEthernet0/0/0]quit
[AR3]
```

AR2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname AR2
[AR2]undo info-center enable
Info: Information center is disabled.
[AR2]interface g0/0/0
[AR2-GigabitEthernet0/0/0]ip address 192.168.1.254 24
[AR2-GigabitEthernet0/0/0]undo shutdown
Info: Interface GigabitEthernet0/0/0 is not shutdown.
[AR2-GigabitEthernet0/0/0]quit
[AR2]inter g0/0/1
[AR2-GigabitEthernet0/0/1]ip address 192.168.2.254 24
[AR2-GigabitEthernet0/0/1]undo shutdown
Info: Interface GigabitEthernet0/0/1 is not shutdown.
[AR2-GigabitEthernet0/0/1]quit
[AR2]
```

3. 使用 PING 测试网络的连通性

AR1 → AR2

```
[AR1]ping 192.168.1.254
  PING 192.168.1.254: 56  data bytes, press CTRL_C to break
    Reply from 192.168.1.254: bytes=56 Sequence=1 ttl=255 time=20 ms
    Reply from 192.168.1.254: bytes=56 Sequence=2 ttl=255 time=20 ms
    Reply from 192.168.1.254: bytes=56 Sequence=3 ttl=255 time=20 ms
    Reply from 192.168.1.254: bytes=56 Sequence=4 ttl=255 time=30 ms
    Reply from 192.168.1.254: bytes=56 Sequence=5 ttl=255 time=20 ms

  --- 192.168.1.254 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/22/30 ms

[AR1]
```

AR1 → AR3

```
[AR1]ping 192.168.2.1
  PING 192.168.2.1: 56  data bytes, press CTRL_C to break
    Request time out
    Request time out
    Request time out
    Request time out
    Request time out

  --- 192.168.2.1 ping statistics ---
    5 packet(s) transmitted
    0 packet(s) received
    100.00% packet loss

[AR1]
```

AR3 → AR2

```
<AR3>ping 192.168.2.254
  PING 192.168.2.254: 56  data bytes, press CTRL_C to break
    Reply from 192.168.2.254: bytes=56 Sequence=1 ttl=255 time=10 ms
    Reply from 192.168.2.254: bytes=56 Sequence=2 ttl=255 time=10 ms
```

```
  Reply from 192.168.2.254: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 192.168.2.254: bytes=56 Sequence=4 ttl=255 time=30 ms
  Reply from 192.168.2.254: bytes=56 Sequence=5 ttl=255 time=30 ms

 --- 192.168.2.254 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/22/30 ms

<AR3>
```

4. 开启 Wireshark 抓包，重新使用 PING 测试网络连通性

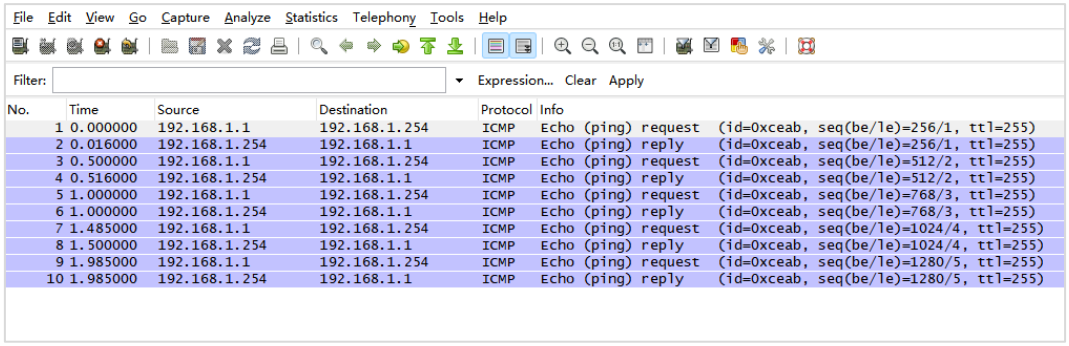


图 2 测试 AR1 到 AR2 的连通性

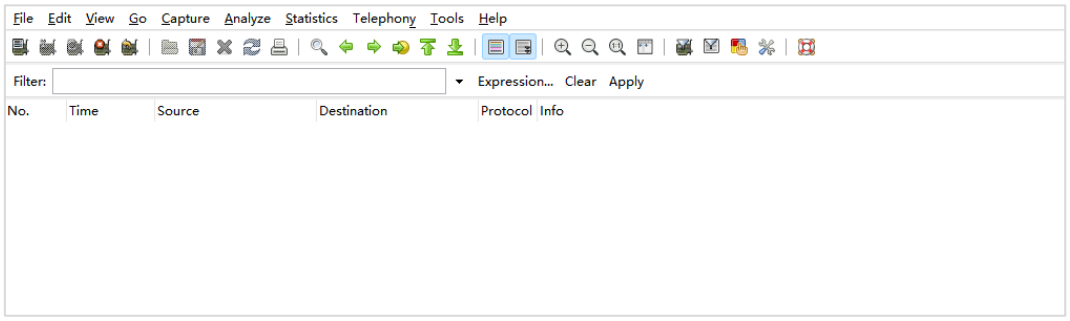


图 3 测试 AR1 到 AR3 的连通性

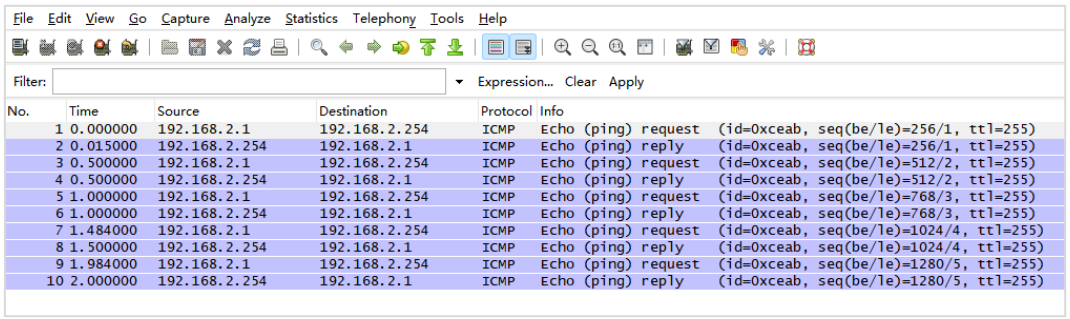


图 4 测试 AR3 到 AR2 的连通性

5. 查看当前路由表

AR1

```
[AR1]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 7        Routes : 7

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
    192.168.1.0/24  Direct  0    0           D   192.168.1.1     GigabitEthernet
0/0/0
    192.168.1.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
  192.168.1.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

AR3

```
<AR3>dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 7        Routes : 7

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0           D   127.0.0.1       InLoopBack0
    192.168.2.0/24  Direct  0    0           D   192.168.2.1     GigabitEthernet
0/0/0
    192.168.2.1/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
  192.168.2.255/32  Direct  0    0           D   127.0.0.1       GigabitEthernet
0/0/0
```

```
255.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0
```

- 127.x.x.x 和 255.255.255.255 是系统自带的本地和广播路由
- R：Relay；中继/迭代；路由查找
- D：Download to FIB；已下发到硬件转发表；路由转发；可以使用

6. 配置静态路由

AR1

```
[AR1]ip route-static 192.168.2.0 24 192.168.1.254
[AR1]
```

- 如果这个时候，AR1 PING AR3，可以成功吗？
- 使用抓包去验证你的结论

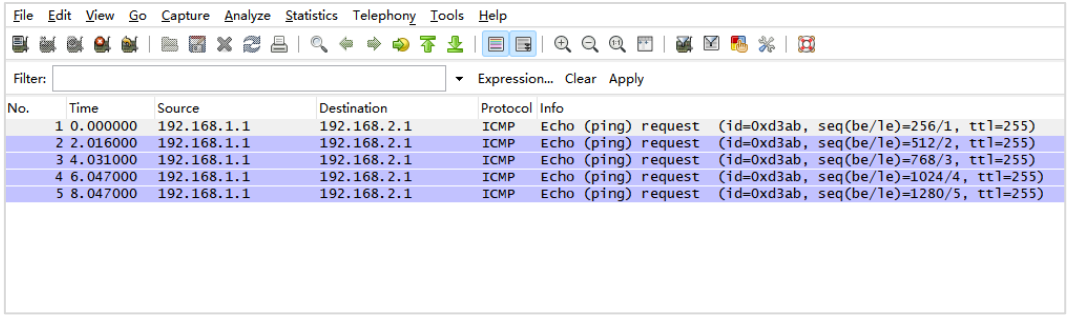


图 5 单侧配置静态路由

AR3

```
[AR3]ip route-static 192.168.1.0 24 192.168.2.254
[AR3]
```

7. 查看当前路由表

AR1

```
[AR1]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0           D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0           D   127.0.0.1       InLoopBack0
```

```
127.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0
   192.168.1.0/24  Direct  0    0          D   192.168.1.1     GigabitEthernet
0/0/0
   192.168.1.1/32  Direct  0    0          D   127.0.0.1       GigabitEthernet
0/0/0
  192.168.1.255/32  Direct  0    0          D   127.0.0.1       GigabitEthernet
0/0/0
   192.168.2.0/24  Static  60   0         RD   192.168.1.254   GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0

[AR1]
```

AR3

```
[AR3]dis ip routing-table
Route Flags: R - relay, D - download to fib
------------------------------------------------------------------------------
Routing Tables: Public
         Destinations : 8        Routes : 8

Destination/Mask    Proto   Pre  Cost      Flags NextHop         Interface

      127.0.0.0/8   Direct  0    0          D   127.0.0.1       InLoopBack0
      127.0.0.1/32  Direct  0    0          D   127.0.0.1       InLoopBack0
127.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0
   192.168.1.0/24  Static  60   0         RD   192.168.2.254   GigabitEthernet
0/0/0
   192.168.2.0/24  Direct  0    0          D   192.168.2.1     GigabitEthernet
0/0/0
   192.168.2.1/32  Direct  0    0          D   127.0.0.1       GigabitEthernet
0/0/0
  192.168.2.255/32  Direct  0    0          D   127.0.0.1       GigabitEthernet
0/0/0
255.255.255.255/32  Direct  0    0          D   127.0.0.1       InLoopBack0

[AR3]
```

8. 测试网络连通性

AR1 → AR3

```
<AR1>ping 192.168.2.1
  PING 192.168.2.1: 56  data bytes, press CTRL_C to break
    Reply from 192.168.2.1: bytes=56 Sequence=1 ttl=254 time=50 ms
    Reply from 192.168.2.1: bytes=56 Sequence=2 ttl=254 time=30 ms
    Reply from 192.168.2.1: bytes=56 Sequence=3 ttl=254 time=30 ms
    Reply from 192.168.2.1: bytes=56 Sequence=4 ttl=254 time=30 ms
    Reply from 192.168.2.1: bytes=56 Sequence=5 ttl=254 time=30 ms

  --- 192.168.2.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 30/34/50 ms

<AR1>
```

AR3 → AR1

```
[AR3]ping 192.168.1.1
  PING 192.168.1.1: 56  data bytes, press CTRL_C to break
    Reply from 192.168.1.1: bytes=56 Sequence=1 ttl=254 time=20 ms
    Reply from 192.168.1.1: bytes=56 Sequence=2 ttl=254 time=50 ms
    Reply from 192.168.1.1: bytes=56 Sequence=3 ttl=254 time=30 ms
    Reply from 192.168.1.1: bytes=56 Sequence=4 ttl=254 time=40 ms
    Reply from 192.168.1.1: bytes=56 Sequence=5 ttl=254 time=30 ms

  --- 192.168.1.1 ping statistics ---
    5 packet(s) transmitted
    5 packet(s) received
    0.00% packet loss
    round-trip min/avg/max = 20/34/50 ms

[AR3]
```

9. 其它测试

拓展练习

- 加入交换机 S5700 和终端 PC
- 调整/增加静态路由/默认路由
- 自己分配网段

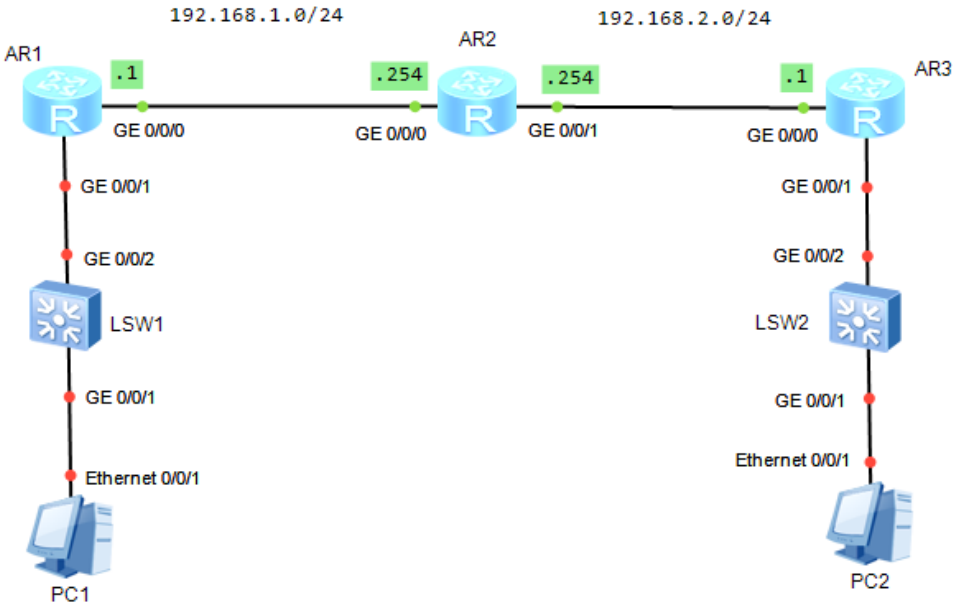


图 6 拓展练习

实验分析

实验总结